PREM TIWARI

Software Engineer II · Backend & Data Engineering

Bengaluru, Karnataka, India • +91 96448 46577 • sepremtiwari@gmail.com

Portfolio • linkedin.com/in/premtiwari14f • github.com/sepkt • leetcode.com/u/prem14f

PROFESSIONAL SUMMARY

Software Engineer II with 3.5+ years building scalable data platforms and cloud-native backend systems — real-time streaming, high-throughput pipelines, CDC, and observability at scale. Strong across Python, PySpark, Spark, Kafka, Flink, Databricks, Snowflake, and AWS, with a focus on cost, reliability, and throughput. Passionate about clean architecture, automation, and building efficient, maintainable systems.

EDUCATION

B.Tech, Information Technology — Jabalpur Engineering College, Jabalpur, M.P. — Jul 2019 – Apr 2023

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL; OOP, DSA, LLD/HLD

Data Engineering: Spark, PySpark, Kafka, Flink, Databricks, Airflow, Airbyte, Snowflake, Delta Lake, DBT, ETL, Data Warehousing, Data Modeling, CDC

AWS & Cloud: EC2, S3, Lambda, Kinesis, MSK, SNS, SQS, Glue, API Gateway, VPC, Route 53, CloudWatch, OpenSearch, Logstash

Databases: PostgreSQL, MongoDB, DynamoDB, Redis, Snowflake

DevOps & Tools: Docker, Kubernetes, KEDA, Terraform, Terragrunt, Ansible, Jenkins, GitHub Actions, Grafana, MLflow, Postman, SonarQube

PROFESSIONAL EXPERIENCE

Software Engineer II — *Smartsheet* — Oct 2025 – Present

- Migrated multi-task streaming jobs from all-purpose to job clusters, cutting compute costs by **$200K+ annually**.
- Re-architected legacy Snowflake pipelines onto Databricks **from scratch** — a reusable, config-driven template ingesting from Airbyte, REST APIs, and Databricks connectors as parallel tasks in one multi-task job, feeding a medallion pipeline (S3 → Auto Loader → Spark Structured Streaming → Gold).
- Built a **CDC pipeline** capturing change-data-capture from the PostgreSQL binlog using **Flink**, streaming into Kinesis (one shard per table) and landing into Delta tables via medallion architecture.
- Drove CI/CD adoption via Databricks Asset Bundles, enabling automated, consistent deployments across environments.
- Migrated **1900+ files** into a new parent folder with **zero breaking changes**, updating CI runners, GitLab CI, and AWS Secrets Manager fetching.
- Built a high-availability log pipeline (CloudWatch → Kinesis → Logstash → OpenSearch) with **sub-second search** and **zero-data-loss** ingestion at scale.
- Hardened legacy ingestion (Airbyte → S3 → Airflow → Snowflake), reducing daily failures by **~34%**.

Data Engineer — *Vestas* — Feb 2024 – Sep 2025

- Built **from scratch** a real-time, **millisecond-latency** alarm platform on Kafka, Redis, and Kinesis (containerized with Docker), exposing an API serving alarms (warning, error, operational) from DynamoDB with event open/close tracking and **Protobuf-encoded payloads**; also streamed alarm data via Kinesis into Delta tables for analytics and data science teams.
- Built MQTT ingestion pulling from a broker and pushing to MSK using async Python libraries, fully containerized in Docker.

- Built an OPC-UA IoT pipeline **from scratch** after an AWS Greengrass POC fell short — asynchronously traversing the OPC-UA node tree, handling encryption, and streaming to AWS MSK, with local-server buffering for resilience; enhanced data availability by **96%**.
- Re-built a legacy Databricks streaming job as a Dockerized service (MSK → Kinesis) with **KEDA autoscaling** on Kafka lag, cutting ingestion latency from minutes to **10–20 seconds**.
- Built a **KEDA-scalable** service replicating data from prod Kafka to QA/Dev Kafka with configurable filtering and row limits, keeping non-prod costs low.
- Optimized Databricks/PySpark pipelines for solar & wind farm data, cutting processing time and latency by **~62%** by switching the source from S3 to Kinesis.
- Built a file download API (DocumentDB metadata + AWS S3 retrieval) with cursor-based pagination across web views listing hundreds of files.

Assistant Data Engineer *— Utopus Insights (a Vestas company)* Feb 2023 – Jan 2024

- Built new-customer ingestion tracking wind-turbine events from API data (warning, error, running, service) with accurate logging; authored LLD & HLD.
- Built **from scratch** a new API data source with **checkpointing** (persisting the last-polled timestamp for gap-free resumption), pushing real-time data into Delta tables; increased data-collection efficiency by **100%**.
- Built a monitoring system surfacing data latency across source services and pinpointing bottlenecks; reduced API latency by **36%**.
- Automated the previously manual, accuracy-based ML model registration to AWS S3, saving the team **12 hrs/week**; raised unit-test coverage to **95%**.
- Delivered seamless Databricks **LTS** upgrades across **50+** jobs with rigorous **data-quality testing** and **zero pipeline failures**.

CERTIFICATIONS

- **Snowflake:** Collaboration & Marketplace, Data Application, Data Lake, Data Warehousing, Data Engineering

ACHIEVEMENTS

- **CodeChef:** 5-star, Global Rank 8, College Rank 1 (Starters 70)
- **LeetCode & GeeksforGeeks:** 450+ problems solved
- **TCS CodeVita:** Round 2 Global Rank 83
- **GFG Job-A-Thon 6:** Global Rank 76